CHAPTER 7 · CONCEPT 3 · WORKSHEET

Practice — Interparticle Spacing, Compressibility and Diffusion

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 · MCQ **[1]**

Interparticle spacing is largest in —

(a) solids (b) liquids (c) gases (d) all equal

Q2 · MCQ **[1]**

Which can be compressed most easily?

(a) a gas (b) a liquid (c) a solid (d) none

Q3 · MCQ **[1]**

The spreading of particles of one substance into another on their own is called —

(a) compression (b) diffusion (c) melting (d) evaporation

Q4 · MCQ **[1]**

Diffusion is fastest in —

(a) solids (b) liquids (c) gases (d) all the same

Q5 · ASSERTION–REASON **[1]**

A: Diffusion becomes faster on heating.
R: Heating makes particles move faster.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK **[1]**

Gases can be _________ easily because of large spaces between their particles.

____

Q7 · FILL IN THE BLANK **[1]**

Diffusion happens because particles are constantly _________.

____

Short answer (2 marks each)

Q8 **[2]**

Arrange solids, liquids and gases in order of increasing interparticle spacing.

Q9 **[2]**

Why are gases easily compressible while liquids are not?

Q10 **[2]**

What is diffusion? In which state is it fastest?

Long / application (3 marks each)

Q11 **[3]**

Using interparticle spacing, explain (i) why a gas fills its whole container and (ii) why we can smell food cooking in another room.

Q12 **[3]**

Describe a syringe activity that shows gases are compressible but liquids are not, and explain the result.

Check yourself → Answer key